D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

Discuss the rationale of combining checkpoint inhibitor (CPI) and VEGF inhibitors in metastatic renal cell carcinoma. Discuss the various CPI with VEGF inhibitor combinations and the evidence. Discuss 2nd line treatment of metastatic renal cell carcinoma.

August 2022 | DM 0822 | Q2 (DEEP Essay / Elaborate (FOCUS 2022–2026)) | **15 marks** | Target ~16 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

0. How to use this DEEP answer

Sections below are **high-yield, trial-dense content** written for a 15-mark Paper IV elaboration. After the deep core, additional revision modules (algorithms, supportive care, flash cards) extend the note to full booklet length.

EXAM PEARLS / HALLMARKS

- In the exam hall, prioritise the DEEP trial tables + NCCN 2026 boxes first.
- Draw at least one flowchart from the deep sections.

1. Introduction — dual targeting in metastatic RCC

Clear-cell RCC is a VEGF-driven, immunogenic cancer. Front-line standards shifted from TKI monotherapy (sunitinib/pazopanib era) to **checkpoint inhibitor (CPI) + VEGF-targeted therapy** doublets and to dual CPI (nivolumab + ipilimumab). This essay covers biologic rationale, landmark combinations with numbers, dosing, toxicity trade-offs, COSMIC-313 nuance, subsequent therapy, and **NCCN Kidney Cancer v2.2027** (ipi+nivo Cat 1).

- Risk-stratify with IMDC (Heng) criteria before choosing 1L.
- Confirm histology (clear-cell vs nccRCC) — most phase III doublets enrolled predominantly ccRCC.

EXAM PEARLS / HALLMARKS

- Examiners want: rationale paragraph + 4 trial names with PFS/OS + 2L algorithm.

2. Biologic rationale — why CPI + VEGF

VHL loss → HIF accumulation → VEGF/PDGF-driven angiogenesis creates an abnormal immunosuppressive tumor microenvironment (TME): MDSCs, regulatory T cells, endothelial barriers to T-cell trafficking, and high VEGF itself impairing dendritic cell function.

VEGF/VEGFR blockade normalizes vessels, reduces MDSCs/Tregs, and can enhance CPI efficacy. PD-1/PD-L1 or CTLA-4 blockade restores antitumor T-cell activity. Combination aims for higher ORR/CR and deeper durable responses than TKI alone.

- Rationale also explains overlapping toxicities (hypertension + immune hepatitis rare but serious).
- HIF-2α inhibitor belzutifan exploits the same VHL–HIF axis downstream — relevant for later lines.

VHL loss / HIF	VEGF-driven angiogenesis
VEGF high TME	MDSC/Treg recruitment; DC dysfunction
Anti-VEGF/TKI	Vessel normalization; ↓ immunosuppression
Anti-PD-1 / CTLA-4	Reinvigorate T-cell cytotoxicity
Clinical result	Higher ORR, longer PFS/OS vs sunitinib

Figure: mechanism / pathway map

3. IMDC risk & choosing among 1L options

IMDC factors: time from diagnosis to systemic therapy <1 year, KPS <80, Hb ULN, neutrophils >ULN, platelets >ULN. Favorable (0), intermediate (1–2), poor (≥3).

Dual CPI (nivo+ipi) historically shone in intermediate/poor risk with durable CR; IO+TKI doublets show high ORR/PFS across risks including favorable. Individualize by need for rapid cytoreduction (IO+TKI), autoimmune history, and preference for treatment-free intervals after limited ipi doses.

1L metastatic ccRCC decision skeleton

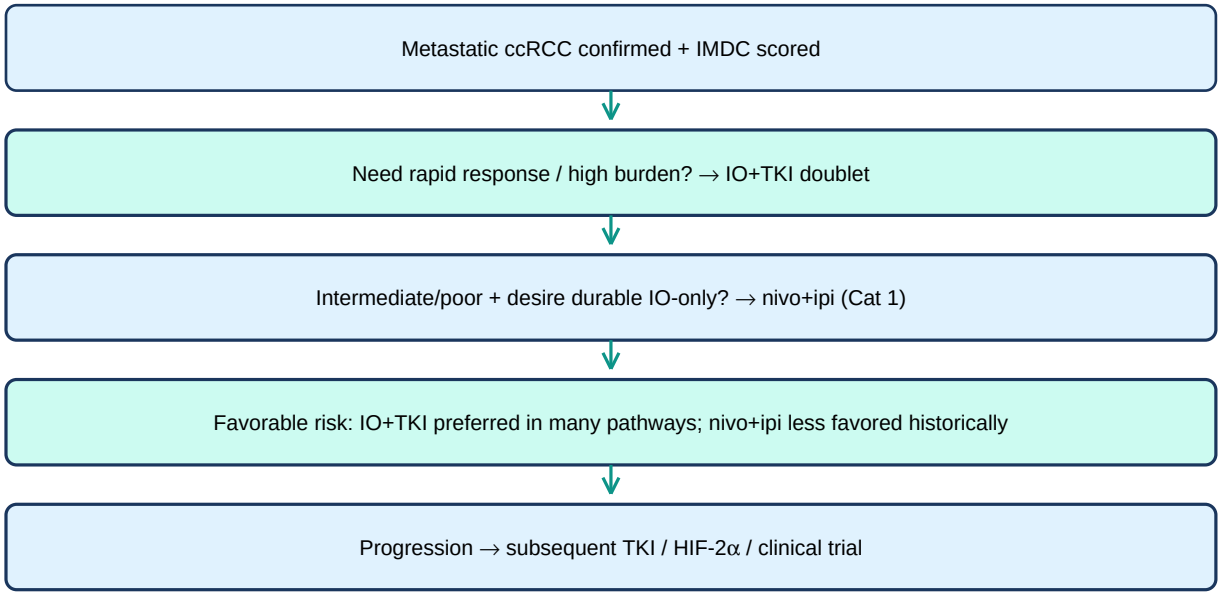


Figure: stepwise exam algorithm

4. CheckMate 214 — nivolumab + ipilimumab

Phase III: nivo 3 mg/kg + ipi 1 mg/kg q3w ×4 → nivolumab maintenance vs sunitinib 50 mg (4/2). Primary focus intermediate/poor IMDC: OS benefit (updated mOS ~47 vs ~27 months range in long follow-up reports; HR ~0.68), ORR ~42% vs ~27%, CR ~10–12% vs ~2%.

Favorable-risk subgroup initially favored sunitinib for ORR/PFS; longer follow-up softens OS differences — still, many guidelines preferred IO+TKI for favorable risk. **NCCN 2026/2027**: ipi+nivo upgraded to **Category 1 preferred** 1L clear-cell RCC — must quote.

Trial	Key result / exam takeaway
CheckMate 214	Nivo+ipi vs sunitinib — OS benefit IMDC int/poor; durable CRs
CheckMate 214 long FU	Sustained OS; plateau suggests cure-like subset

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Nivolumab + ipilimumab induction	Nivo 3 mg/kg + ipi 1 mg/kg IV q3w ×4
Nivolumab maintenance	240 mg q2w or 480 mg q4w (label-aligned)
Sunitinib (control)	50 mg PO daily 4 weeks on / 2 off

Table: doses and regimens

EXAM PEARLS / HALLMARKS

• Ipi dose in RCC is 1 mg/kg (lower than melanoma 3 mg/kg combo) — fewer grade 3–4 irAEs than melanoma dosing but still significant.

5. KEYNOTE-426 — pembrolizumab + axitinib

Pembro 200 mg q3w + axitinib 5 mg BID vs sunitinib. Landmark: improved OS and PFS across IMDC groups. Illustrative numbers: mPFS ~15.4 vs 11.1 months; ORR ~60% vs ~40%; OS HR ~0.73 with mOS ~45–47 vs ~40 months in updates.

Axitinib is a selective VEGFR TKI — hypertension and diarrhea common; hold axi for immune hepatitis workup overlap.

Trial	Key result / exam takeaway
KEYNOTE-426	Pembro+axi vs sunitinib — OS/PFS/ORR superiority

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV q3w or 400 mg q6w
Axitinib	Start 5 mg PO BID; titrate 2.5–10 mg BID per BP/toxicity

Table: doses and regimens

6. CLEAR — lenvatinib + pembrolizumab

Lenvatinib 20 mg daily + pembrolizumab vs sunitinib. Among the highest ORR/PFS of IO+TKI triplets-in-spirit doublets: mPFS ~23.9 vs 9.2 months; ORR ~71% vs ~36%; OS HR ~0.66 in primary/updated analyses.

Toxicity: hypertension, diarrhea, proteinuria, dysphonia, weight loss — proactive dose reduction of lenvatinib common and acceptable.

Trial	Key result / exam takeaway
CLEAR (KEYNOTE-581)	Len+pembro vs sunitinib — mPFS 23.9 vs 9.2 mo; high ORR

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Lenvatinib (with pembro)	20 mg PO daily
Pembrolizumab	200 mg q3w
Lenvatinib dose reductions	14 → 10 → 8 mg daily per label toxicity path

Table: doses and regimens

EXAM PEARLS / HALLMARKS

• Quote 23.9 vs 9.2 months PFS — highest-yield CLEAR number.

7. CheckMate 9ER — cabozantinib + nivolumab

Cabozantinib 40 mg daily + nivolumab vs sunitinib. Cabo hits VEGFR/MET/AXL — useful biology after VEGF resistance. mPFS ~16.6 vs 8.3 months; ORR ~56% vs 27%; OS HR ~0.70.

Cabo toxicity: diarrhea, PPE, hypertension, mucositis; start 40 mg (not 60 mg monotherapy dose) in the IO combo.

Trial	Key result / exam takeaway
CheckMate 9ER	Cabo 40 mg + nivo vs sunitinib — PFS/OS/ORR benefit

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Cabozantinib (with nivo)	40 mg PO daily empty stomach
Nivolumab	240 mg q2w or 480 mg q4w

Table: doses and regimens

8. Other IO+VEGF / dual IO references

JAVELIN Renal 101 (avelumab+axitinib) improved PFS; OS maturity less compelling than pembro-based doublets — NCCN demotions occurred over time. NCCN 2026 notes axitinib + avelumab removed and some post-IO TKI+IO combinations downgraded — show currency.

- Do not conflate adjuvant pembro with metastatic doublets.
- PD-L1 is not required to select 1L IO in ccRCC.

Trial	Key result / exam takeaway
JAVELIN Renal 101	Avelumab+axi — PFS benefit; OS less practice-defining
KEYNOTE-564	Adjuvant pembrolizumab post-nephrectomy high-risk — separate from metastatic but cite literacy

Table: landmark clinical trials

9. COSMIC-313 nuance — triplet not automatic standard

COSMIC-313: cabozantinib + nivo + ipi vs nivo + ipi in IMDC intermediate/poor. Added cabo improved PFS but with substantially more grade 3–4 toxicity (ALT/AST, diarrhea, PPE). OS advantage not established as practice-changing at key analyses — therefore triplet is **not** default 1L for all; discuss only in selected high-risk patients or trials.

- Exam maturity: 'PFS≠adopt triplet blindly; toxicity and OS matter.'
- Know why doublet remains standard.

Trial	Key result / exam takeaway
COSMIC-313	Cabo+nivo+ipi vs nivo+ipi — PFS↑, toxicity↑, OS unclear/not standard for all

Table: landmark clinical trials

10. Toxicity management of CPI + VEGF doublets

VEGF side effects: hypertension, proteinuria, bleeding/clotting, wound-healing delay (hold perioperatively), PPE, dysphonia, diarrhea. CPI: irAEs as usual. Overlap: hepatitis, diarrhea — hold both agents, diagnose immune vs TKI etiology (timing, enzyme pattern), use steroids if immune.

- Home BP monitoring mandatory on axi/len/cabo.
- Educate for colitis/pneumonitis red flags.

Toxicity triage on IO+TKI

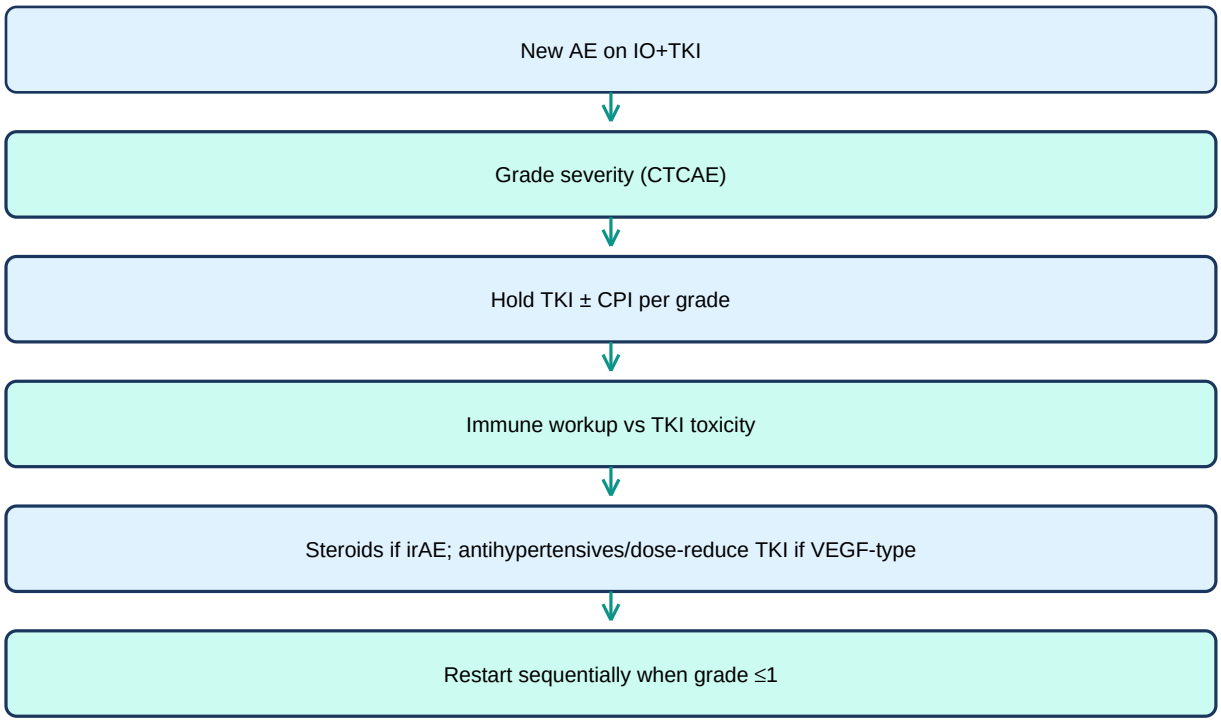


Figure: stepwise exam algorithm

11. Second-line and subsequent therapy after IO ± VEGF

After IO+TKI progression: cabozantinib (if not used), lenvatinib + everolimus, tivozanib (TIVO-3), other VEGF TKIs (axitinib, sorafenib historically), belzutifan (HIF-2α) for VHL-associated and selected sporadic ccRCC after prior PD-1 and VEGF therapy.

After nivo+ipi progression: VEGF TKI monotherapy or IO+TKI if not given — sequencing evidence evolving. Rechallenge with CPI alone less established. Cytoreductive nephrectomy selectively in IMDC favorable synchronous disease (CARMENA caution).

Subsequent-line mRCC after modern 1L

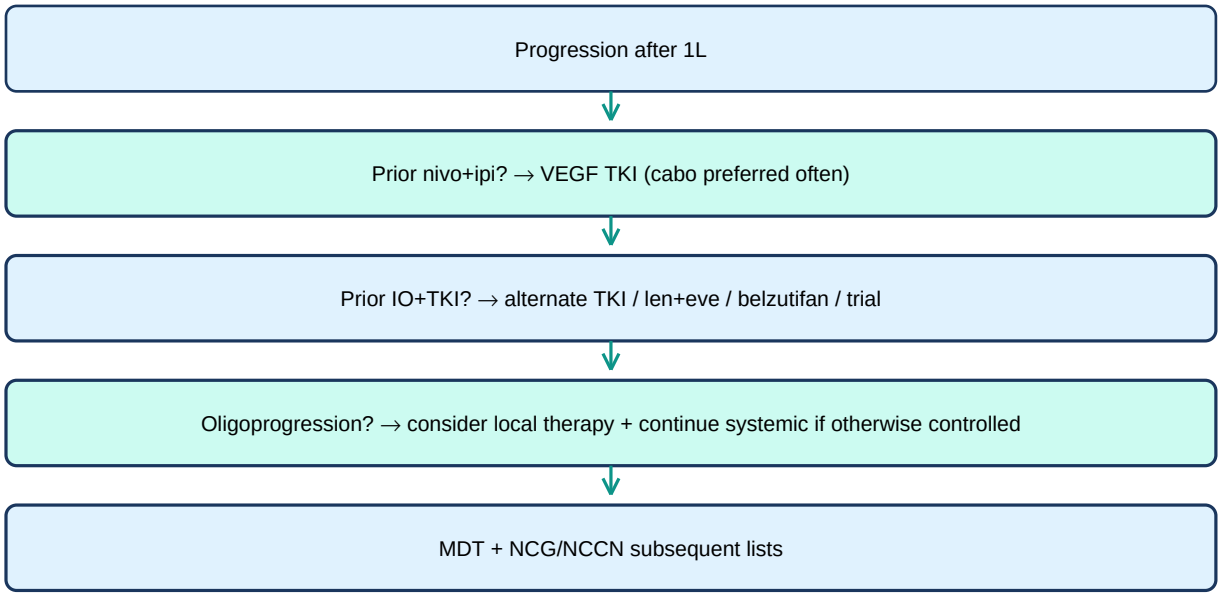


Figure: stepwise exam algorithm

Trial	Key result / exam takeaway
METEOR	Cabozantinib vs everolimus post-VEGF — OS/PFS benefit
CheckMate 025	Nivo vs everolimus — historical 2L IO
TIVO-3	Tivozanib vs sorafenib — PFS in highly pretreated
LEN+EVE (Study 205 / approvals)	Len 18 mg + everolimus 5 mg after VEGF
Belzutifan trials (LITESPARK program)	HIF-2α inhibitor activity post IO/VEGF

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Cabozantinib monotherapy	60 mg PO daily (empty stomach)
Lenvatinib + everolimus	18 mg + 5 mg PO daily
Tivozanib	1.34 mg daily 21/7 schedule per label
Belzutifan	120 mg PO daily (monitor Hb — anemia)

Table: doses and regimens

12. Non–clear cell & special populations

nccRCC: heterogeneous (papillary, chromophobe, translocation, collecting duct). Evidence thinner; cabo, immuno-oncology, and genomic matching (MET, FH, SWI/SNF) matter. NCCN 2026: consider tumor genomic testing for trial eligibility.

Brain mets: local therapy + systemic with CNS activity considerations. Poor PS: prefer manageable TKI or best supportive care.

- Sarcomatoid features: IO-containing regimens preferred (CheckMate 214 sarcomatoid benefit notable).
- Pregnancy: avoid TKI/IO — counsel.

13. NCCN Kidney Cancer v2.2027 / 2026 updates (must-box)

Paper IV marks for explicitly writing NCCN category changes.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Kidney Cancer v2.2027; NCCN Immunotherapy toxicity guidelines (supportive)
- Ipilimumab + nivolumab upgraded to Category 1 preferred 1L clear-cell RCC.
- Lenvatinib Cat 2A useful in certain circumstances subsequent (IO-naive or post-IO).
- Several post-IO TKI+IO combinations downgraded Cat 2A→2B; axitinib + avelumab removed.
- Non–clear cell: consider tumor genomic testing for trial eligibility.
- Subsequent therapy: prefer agents with OS/PFS data not previously used (cabo, len+eve, tivozanib, belzutifan as indicated).

14. Indian / NCG practical notes

Cost and access: sunitinib/pazopanib generics still used when IO unavailable — disclose resource-stratified care. IO+TKI and nivo+ipi available at tertiary/NCG centers via schemes; biosimilars limited for CPI. Monitor BP in clinic networks; educate on irAE in local language.

- If only TKI accessible: sunitinib/pazopanib/cabo still valid — plan IO referral when feasible.
- Cytoreductive nephrectomy not routine for poor-risk IMDC (CARMENA).

15. Comparison table for the answer book

Draw a 4-row table in the hall: regimen | key trial | mPFS highlight | OS/ORR pearl | top toxicity.

- Column for IMDC applicability.
- Column for whether CTLA-4 included (irAE burden).

Trial	Key result / exam takeaway
Nivo+ipi	CM-214 — OS int/poor; CR ~10%+; Cat 1 preferred
Pembro+axi	KN-426 — OS benefit; ORR ~60%
Len+pembro	CLEAR — mPFS 23.9 vs 9.2; ORR ~71%
Cabo+nivo	9ER — mPFS 16.6 vs 8.3; cabo 40 mg
Cabo+nivo+ipi	COSMIC-313 — PFS↑ only; not default

Table: landmark clinical trials

16. Closing summary & exam pearls

CPI + VEGF combinations exploit angiogenesis–immune crosstalk in ccRCC and outperform sunitinib on ORR/PFS/OS. Choose among pembro+axi, len+pembro, cabo+nivo, and nivo+ipi using IMDC, disease tempo, and toxicity. COSMIC-313 shows more is not always better. Subsequent therapy relies on unused VEGF TKIs, len+everolimus, tivozanib, and HIF-2α inhibition per NCCN 2026/2027.

EXAM PEARLS / HALLMARKS

- Rationale: VEGF shapes immunosuppressive TME — dual blockade synergistic.
- CLEAR PFS 23.9 vs 9.2; 9ER 16.6 vs 8.3; KN-426 OS HR ~0.73.
- CM-214: nivo 3 + ipi 1 ×4 → nivo; Cat 1 preferred NCCN.
- Cabo with nivo is 40 mg — not 60 mg.
- COSMIC-313: PFS without clear OS mandate for triplet.
- 2L: cabo / len+eve / belzutifan / tivozanib — avoid recycling identical mechanisms blindly.

— Additional exam-length revision modules —

The following modules reinforce algorithms, toxicities, Indian/NCG context, and last-page summary structure.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

Work-up algorithm (draw this in the answer booklet)

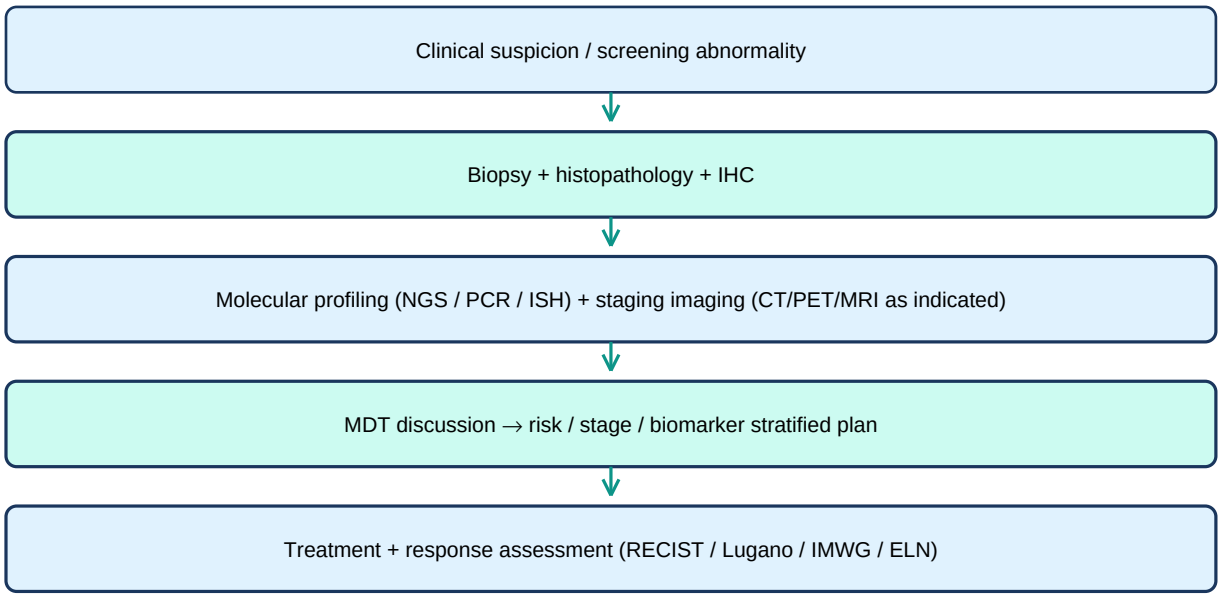


Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.
- Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).
- Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.
- Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).
- Re-biopsy / liquid biopsy at progression to map resistance.

Management flowchart (reproduce in exam)

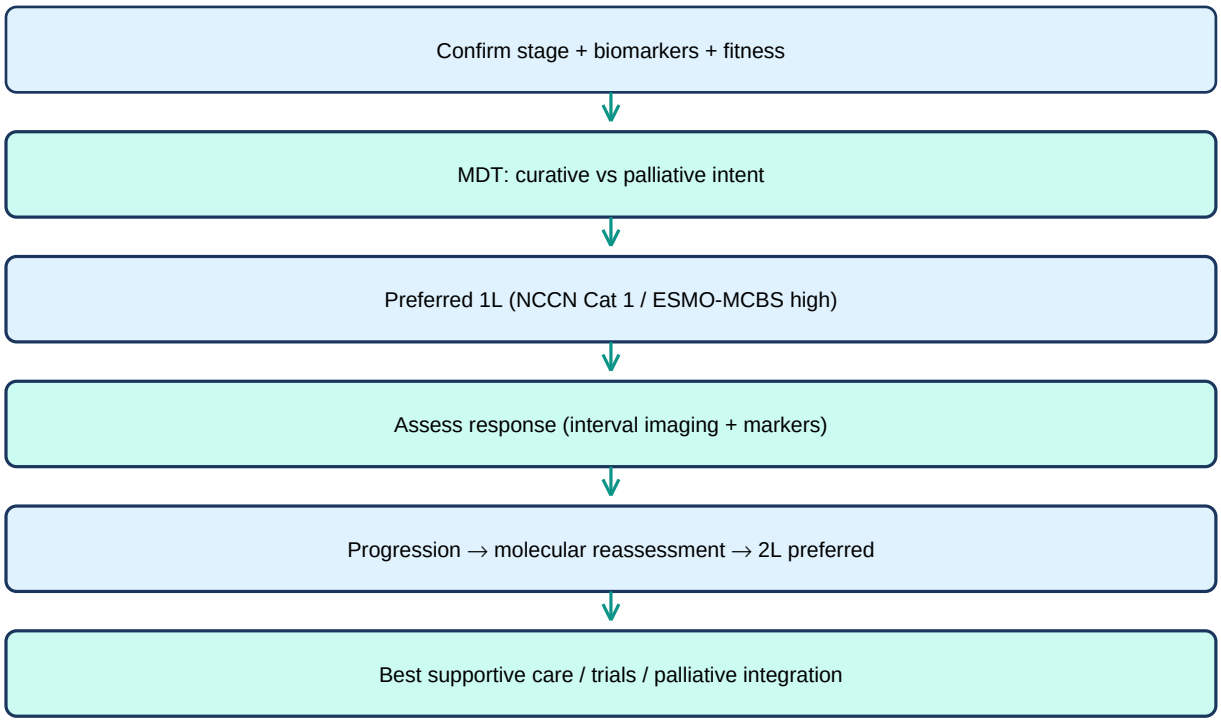


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

- Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.
- Distinguish statistical vs clinical significance; note crossover and subgroup caveats.
- For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
CheckMate 214	Nivo + ipi — now Cat 1 preferred NCCN 2026 ccRCC
KEYNOTE-426 / CLEAR / CheckMate 9ER	Pembrolizumab+axi; len+pembro; cabo+nivo
KEYNOTE-564	Adjuvant pembrolizumab
METEOR / CABOSUN	Cabozantinib

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

- Anticipate class toxicities and counsel patients; use CTCAE grading.

• Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.

• IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.

• ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.

• TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.

• Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

• Algorithm: grade 1 → continue with close watch; grade 2 → hold + steroids; grade 3–4 → discontinue permanently for many toxicities + high-dose steroids ± infliximab/vedolizumab/MMF.

Toxicity management loop

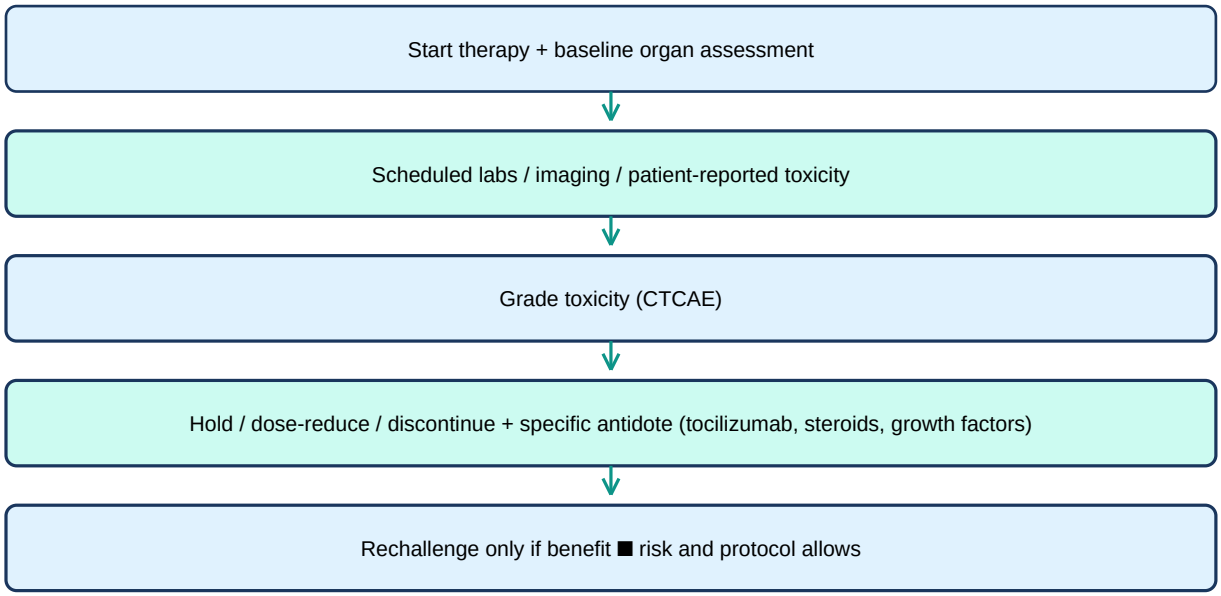


Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for 'recent advances')

Paper IV rewards currency. Explicitly write: **"As per NCCN Guidelines updates in 2026/2027…"** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

• State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.

• If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* Kidney Cancer v2.2027

• Ipilimumab + nivolumab upgraded to Cat 1 preferred 1L clear-cell RCC.

• Lenvatinib Cat 2A useful in certain circumstances subsequent (IO-naive or post-IO).

• Several post-IO TKI+IO combinations downgraded Cat 2A→2B; axitinib + avelumab removed.

• Non–clear cell: consider tumor genomic testing for trial eligibility.

• Earlier-line CPI expansion: ATOMIC stage III dMMR colon; neoadjuvant CPI for dMMR rectal; neoadjuvant melanoma; EV+P bladder; FLOT+durvalumab gastric.

• PD-L1 testing emphasized for gastric/EGJ; limited benefit if PD-L1 negative.

EXAM PEARLS / HALLMARKS

• NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).

• Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

Discuss differentials, mimics, and when to refer to radiation oncology, surgery, palliative care, genetics, or cell-therapy centers.

Include brief notes on health economics (cost-effectiveness, biosimilars) and clinical trial enrollment as standard options.

- MDT documentation improves outcomes and exam completeness.
- Survivorship care plans: late effects, second malignancies, cardiac, endocrine, psychosocial.
- Real-world evidence and digital tools increasingly complement RCT data (relevant to 2026 Paper IV themes).

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

- Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.
- Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.
- Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.
- Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

- Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)
- Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 15-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification
- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

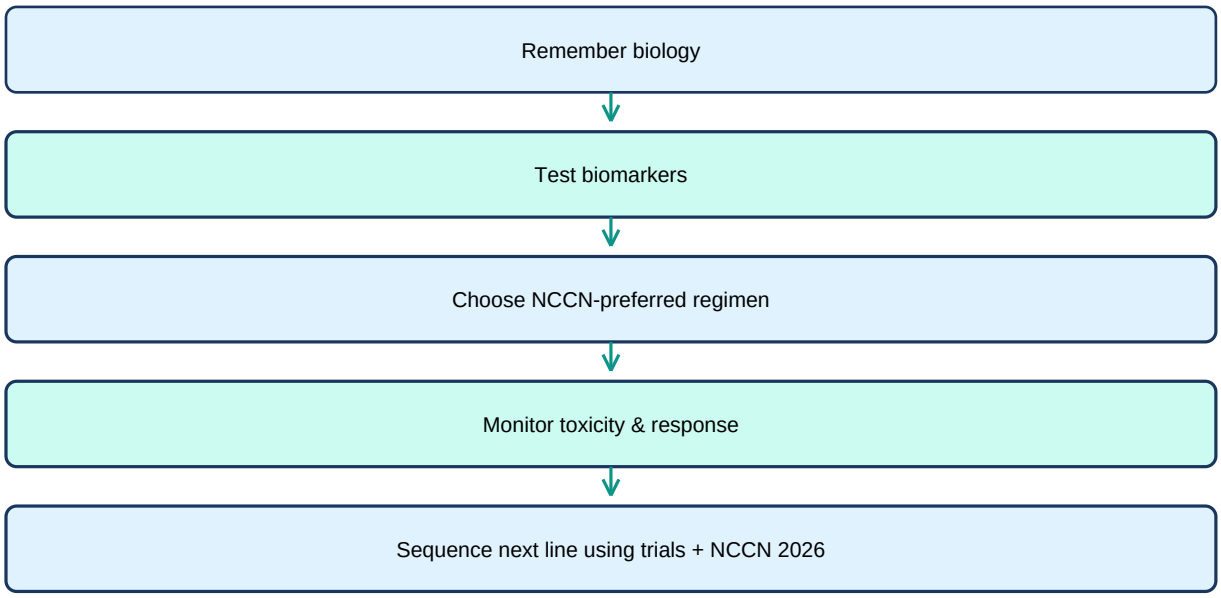


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say 'NCCN 2026' explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.
- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.

• ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

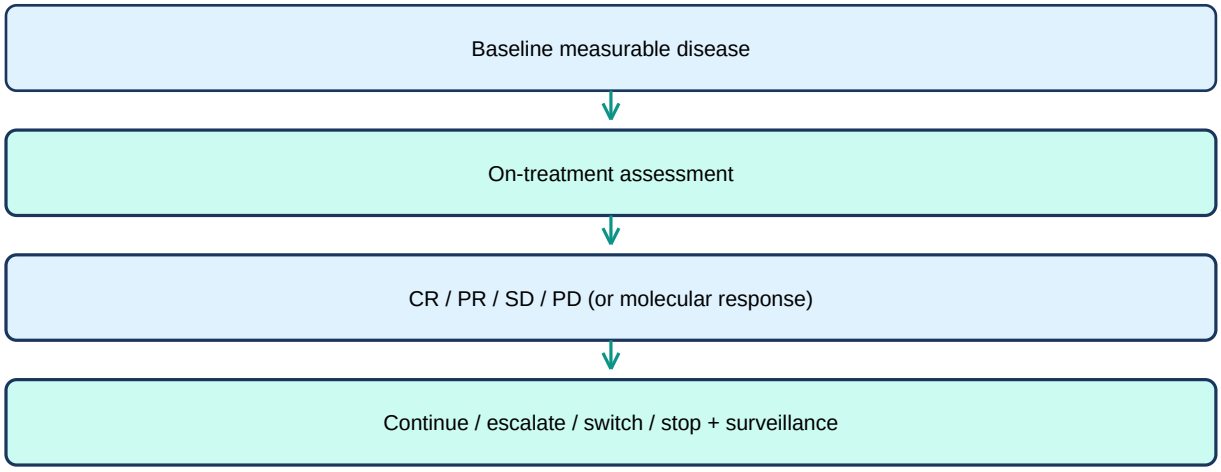


Figure: stepwise exam algorithm

13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

• Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.

• Another table: resistance mutation → preferred next drug.

Trial	Key result / exam takeaway
CheckMate 214	Nivo + ipi — now Cat 1 preferred NCCN 2026 ccRCC
KEYNOTE-426 / CLEAR / CheckMate 9ER	Pembrolizumab+axi; len+pembro; cabo+nivo
KEYNOTE-564	Adjuvant pembrolizumab
METEOR / CABOSUN	Cabozantinib

Table: landmark clinical trials

14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

• Document MDT decisions; clinical trial referral when standard options exhausted.

• Counsel financial toxicity transparently; prioritize highest-value Category 1 options.

• Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

15. Sample ‘examiner hot spots’ & model closing paragraph

Model closing: “In summary, management of this problem is biomarker-driven. Current standards are supported by phase III trials and reflected in NCCN 2026/2027 updates. Sequencing should maximize OS and QoL while mitigating class-specific toxicities. Enrollment in clinical trials and adherence to MDT pathways remain integral.”

Hot spots: exact doses, one HR/median, resistance mechanism, NCCN category, and a drawn algorithm.

EXAM PEARLS / HALLMARKS

• If stuck on memory: write mechanism → class → one trial → one dose → one toxicity → NCCN 2026 line.

• Never contradict yourself between algorithm and trial table.

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

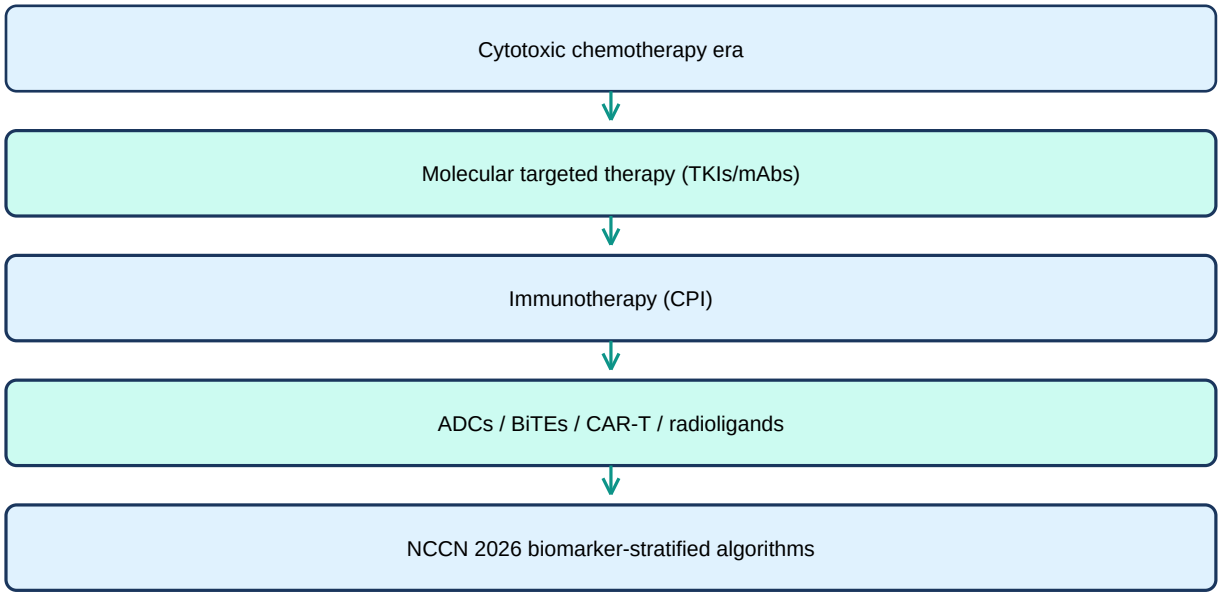


Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

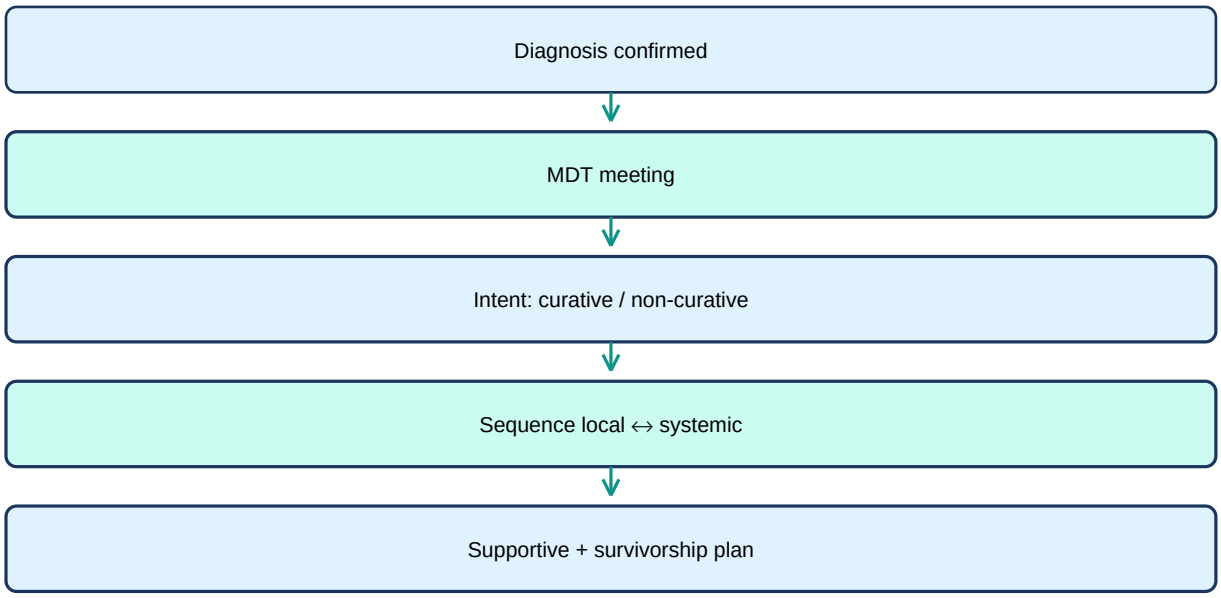


Figure: stepwise exam algorithm

E. Immunotherapy deep notes

CTLA-4 acts priming phase in lymphoid organs; PD-1/PD-L1 in peripheral effector phase. LAG-3 (relatlimab) additional checkpoint.

Biomarkers: PD-L1, MSI/dMMR, TMB — imperfect. TME, gut microbiome modulation is an exam discussion point (2026 short note theme).

Neoadjuvant IO preferred for resectable nodal melanoma (NCCN 2026). Combinations with VEGF in RCC; with chemo in lung/gastric; with ADC in bladder (EV+P).

- irAE timelines: skin/GI early; endocrine anytime; late neuropathies/myocarditis rare but deadly.
- Rechallenge decisions individualized after grade 2 resolution; avoid after grade 4 myocarditis/SJS etc.

E. RCC deep notes

Clear cell vs non-clear cell. IMDC risk for 1L choice. NCCN 2026: nivo+ipi Cat 1 preferred; post-IO sequencing recalibrated; genomic testing in nccRCC for trials.

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

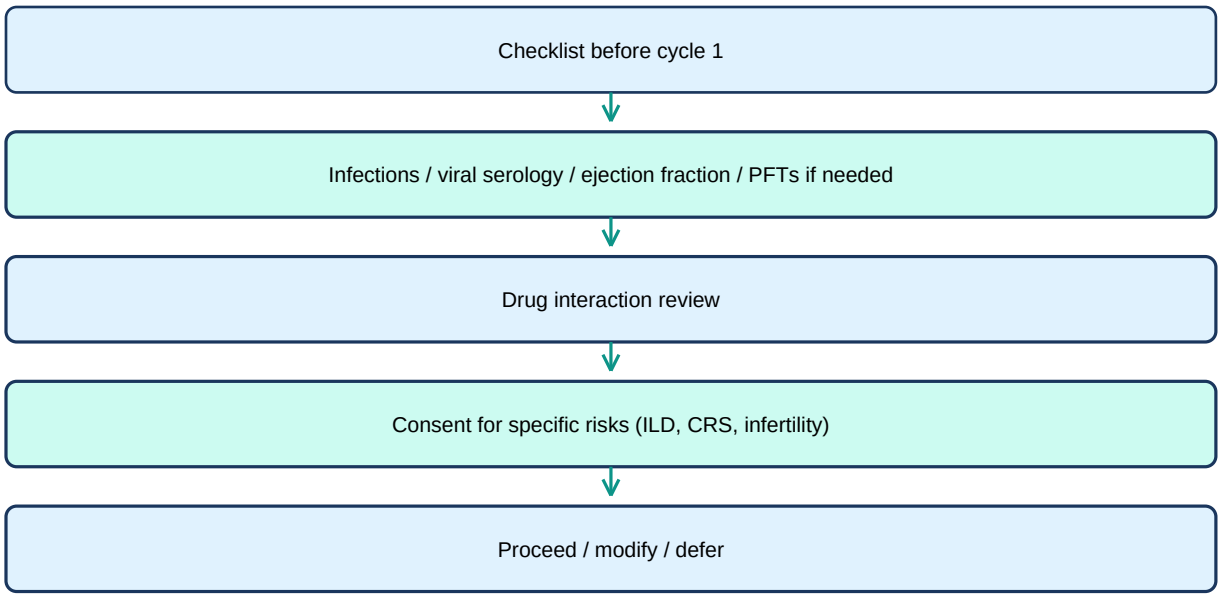


Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

- Wall chart + flowchart + trial table = visual completeness.
- Write ‘NCCN 2026’ in quadrant 4 explicitly.

I2. Full-length essay addendum — tables to copy into the answer book

For 15-mark elaborations, add two hand-drawn tables: (Table A) Drug–dose–pearl; (Table B) Trial–population–endpoint–practice change. Then write a short paragraph on quality-of-life and patient-reported outcomes, which modern trials increasingly report.

Discuss biomarker evolution on therapy (liquid biopsy cadence), and when to stop therapy (duration of CPI, TFR in CML, maintenance stopping rules).

Include a brief note on second malignancies and late effects when curative-intent therapy is discussed (alkylators, RT fields, PARPi MDS/AML risk, CHL survivors).

- Table A columns: Drug | Dose | Key toxicity | Pearl
- Table B columns: Trial | Arm | Primary endpoint | Result | Change in practice
- One paragraph on shared decision-making and financial counseling

Trial	Key result / exam takeaway

CheckMate 214	Nivo + ipi — now Cat 1 preferred NCCN 2026 ccRCC
KEYNOTE-426 / CLEAR / CheckMate 9ER	Pembrolizumab+axi; len+pembro; cabo+nivo
KEYNOTE-564	Adjuvant pembrolizumab
METEOR / CABOSUN	Cabozantinib

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* Kidney Cancer v2.2027

• Ipilimumab + nivolumab upgraded to Cat 1 preferred 1L clear-cell RCC.

• Lenvatinib Cat 2A useful in certain circumstances subsequent (IO-naive or post-IO).

• Several post-IO TKI+IO combinations downgraded Cat 2A→2B; axitinib + avelumab removed.

• Non–clear cell: consider tumor genomic testing for trial eligibility.

• Earlier-line CPI expansion: ATOMIC stage III dMMR colon; neoadjuvant CPI for dMMR rectal; neoadjuvant melanoma; EV+P bladder; FLOT+durvalumab gastric.

• PD-L1 testing emphasized for gastric/EGJ; limited benefit if PD-L1 negative.

I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one ‘if progression then…’ resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway

- Flash 3: Mandatory biomarker
- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn
- Flash 8: 2026 guideline change
- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

EXAM PEARLS / HALLMARKS

- If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.
- Keep handwriting large for doses and HRs.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

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Final NCCN 2026 box

Copy these bullets onto the last page of your answer booklet for Paper IV currency marks.

NCCN 2026 / 2027 UPDATES (include in answer)

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Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.